$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \qquad (3.20)$$

These three matrices are very famous and carry the name of their discoverer. They are the *Pauli matrices.*[4]

3.5 A COMMON MISCONCEPTION

This is a convenient time to warn you about a potential hazard. The correspondence between operators and measurements is fundamental in quantum mechanics. It is also very easy to misunderstand. Here's what is true about operators in quantum mechanics:

1. Operators are the things we use to calculate eigenvalues and eigenvectors.

2. Operators act on state-vectors (which are abstract mathematical objects), not on actual physical systems.

3. When an operator acts on a state-vector, it produces a new state-vector.

Having said what is true about operators, I want to warn you about a common misconception. It is often thought that measuring an observable is the same as operating with the corresponding operator on the state. For example, suppose we are interested in measuring an observable **L**. The measurement is some kind of operation that the apparatus does to the